
AN ENSEMBLE APPROACH TO PORTFOLIO OPTIMIZATION

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ABSTRACT

Markowitz' optimization relies on correct estimates of expected returns and covariance matrix. However, a very long time series of returns is required to get reliable sample estimates, and such data is usually unavailable. Literature suggests cleaning either the sample estimates before employing them as inputs or imposing some structure on portfolio weights. We suggest an alternative strategy based on ensemble technique. Our approach blends weights from four popular strategies namely - Volatility Minimisation, Sharpe Maximisation, Geometric Mean Maximisation and Equally-Weighted portfolio. By only employing a structure on the covariance matrix, we find that such a strategy reduces the likelihood of extreme allocation leading to lower variance in weights and hence lower estimation risk. An empirical study using Indian data finds that the suggested strategy exhibits larger Sharpe ratios and smaller drawdown out-of-sample compared to naive equally weighted strategy.

Keywords Portfolio Optimization · Ensemble Portfolio · Shrinkage Estimators

JEL Classification C13, J11, C61

1 Introduction

Pioneering and ground-breaking research in portfolio optimization modelling was conducted by Harry Markowitz in 1952, which earned him the Nobel Prize in Economics. His mean-variance (MV) portfolio optimization is a single-period model which assigns weights to assets within a portfolio to achieve the optimal trade-off between the mean (as a measure of profit) and the variance (as a measure of risk) of the portfolio returns for a future period. For a given portfolio of n assets (say stocks) with expected returns μ_i , let w_i be the weight or share of the portfolio's value invested in asset i , such that $\sum_{i=1}^n w_i = 1$. Further, let $\mathbf{w} = (w_1, \dots, w_n)^T$, $\boldsymbol{\mu} = (\mu_1, \dots, \mu_n)^T$ and $\mathbf{1} = (1, \dots, 1)^T$. Hence, the portfolio return has mean $\mathbf{w}^T \boldsymbol{\mu}$ and variance $\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}$, where $\boldsymbol{\Sigma}$ is the covariance matrix of the asset returns. The MV model optimises the said trade-off by setting a target value for mean returns ($\boldsymbol{\mu}_*$) while minimising risk to derive a vector of optimal weights, $\mathbf{w}_{opt}$; we can see this in the optimisation problem below

$$\begin{aligned} \mathbf{w}_{opt} &= \arg \min_{\mathbf{w}} \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} \\ \text{subject to} \quad & \mathbf{w}^T \boldsymbol{\mu} = \boldsymbol{\mu}_*, \mathbf{w}^T \mathbf{1} = 1, \mathbf{w} \geq \mathbf{0} \end{aligned} \tag{1}$$

Theoretically, Markowitz' allocation stands on firm ground. However, problems arise in its implementation.

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1.1 Optimization as an *error maximizing* exercise

The theory assumes that the expected returns vector (μ) and the covariance matrix (Σ) are known. But in practice, these are usually estimated from historical data under i.i.d. assumptions and suffer from estimation errors. MV optimizers are efficient at exploiting small differences in returns and covariances amongst assets however, this efficiency becomes a curse in the presence of estimation errors (Michaud 1989 [4]). As a result, MV portfolio weights are extremely sensitive to changes in sample estimates (Best and Grauer 1991 [5]). Accounting for parameter uncertainty, for instance MV optimization under Stein-estimates, usually outperforms standard plug-in approach (Chopra et al 1993 [7]).

Generally, two approaches have been suggested to account for estimation risk. The first approach proposes some structure on the moments of asset returns (Jorion 1986 [3]; Ledoit and Wolf 2003 [8]; Kourtis et al. 2012 [18]) while the other proposes a structure on the portfolio weights (Kan and Zhou 2007 [10]; Frahm and Memmel 2010 [13]; Pollak 2011 [16]; Tu and Zhou 2011 [17]).

The objective of this study is to suggest an alternative approach to minimize the risk of both "overfitting" on a strategy as well as "estimation errors" in the sample estimates. First, we use a shrinkage estimate of covariance matrix as proposed by Ledoit and Wolf (2004) [9] to generate an improved estimate for the covariance matrix, Σ . Second, we propose an ensemble approach to asset allocation that blends four allocation strategies - Volatility Minimisation, Sharpe Maximisation, Geometric Mean Maximisation and Equally-Weighted portfolio to generate superior allocations.

The rest of this paper is organised as follows: Section 2 highlights the different approaches to estimating the $\hat{\mu}$ and $\hat{\Sigma}$ with greater focus on the shrinkage approach. Section 3 discusses three variants of the classic mean-variance optimization. Section 4 describes the proposed approach and demonstrates the effectiveness of our approach through an empirical study. Section 5 concludes the study.

2 Covariance matrix estimation

There are a few different approaches to estimating the expected returns vector for example the bootstrapping approach of Michaud (1989) and the quasi-Bayesian approach of Black and Litterman (1992) [6]. In this study we use the sample estimate of the expected returns $\hat{\mu}$ as μ . However, there are no limitations imposed by our approach on an improved estimation of expected returns so that an interested investor can use it to complement the suggested strategy.

As for the covariance matrix, mainly three approaches exist in the literature as pointed out by T Lai et al. (2011) [14]. The first of these approaches estimates the covariance matrix implied by a multi-factor model of stock returns. The second approach uses Bayesian and other shrinkage estimators and the third one is based on bootstrapping to reduce the estimation error. For the purposes of this study, we use the Shrinkage approach of Ledoit and Wolf (2004). This approach is simple and intuitive. We know extreme return realisations can wreak havoc on sample covariance estimators. In the shrinkage approach, we shrink the classical estimators to a *target* based on economic theory. Here, we assume the target covariance matrix to have constant variance and zero covariance.

2.1 Shrinking the covariance matrix

Consider the sample covariance estimate for a $T \times n$ sample to be S , where T is the time series of returns and n is the number of assets. Next, we define a target covariance matrix F to which we want to shrink the sample covariance matrix. F can take different forms as per our theoretical understanding of the long-run relationships between returns. For example, we can assume that there is constant variance and no covariance between asset returns i.e. $f_{i,i} = f_{j,j} = c$, where c is some scaling constant and $f_{i,j} = 0 \quad \forall \quad i, j$ pairs. In this case, F takes the form of cI , where I is an $n \times n$ identity matrix. This approach is suggested by Ledoit-Wolf (2004).

$$\begin{aligned}\hat{\Sigma} &= (1 - \delta)S + \delta F \\ &= (1 - \delta)S + \delta \kappa I\end{aligned}\tag{2}$$

where δ is the shrinkage factor and κ is the average variance of the assets. A nice implementation of the Identity matrix based approach is provided in *scikit-learn*[15] python package.

3 Optimization under the Mean-Variance framework

As explained in Section 1, an individual investor can choose to optimize a number of objective functions based on her preferences or investment objectives. Three main approaches amongst them are given below.

3.1 Volatility minimization

The objective in this approach is to find a portfolio that minimizes variance. This is also referred to as the Global Minimum Variance Portfolio, hereafter referred to as GMV. This strategy is attractive amongst practitioners as we do not need to input an estimate of expected returns for extracting the weights.

$$\begin{aligned}\mathbf{w}_{vol} &= \arg \min_{\mathbf{w}} \mathbf{w}^T \Sigma \mathbf{w} \\ \text{subject to } & \mathbf{w}^T \mathbf{1} = 1, \mathbf{w}_i \geq 0 \text{ for } i \in \{1, 2, \dots, n\}\end{aligned}$$

3.2 Sharpe ratio maximization

The most popular implementation of mean-variance optimization is maximizing risk-adjusted returns which amounts to maximizing a portfolio's Sharpe ratio, where risk is measured by standard deviation; hereafter referred to as SRM.

$$\begin{aligned}\mathbf{w}_{shp} &= \arg \max_{\mathbf{w}} \frac{\mathbf{w}^T \boldsymbol{\mu} - R_f}{(\mathbf{w}^T \Sigma \mathbf{w})^{1/2}} \\ \text{subject to } & \mathbf{w}^T \mathbf{1} = 1, \mathbf{w}_i \geq 0 \text{ for } i \in \{1, 2, \dots, n\}\end{aligned}$$

where R_f is the risk free rate.

3.3 Geometric mean maximization

Although Geometric mean maximization (GMM) is not strictly a Mean-Variance optimization strategy and is incompatible with the expected utility framework, it is nevertheless based on an equally appealing idea. It aims at maximizing the expected growth rate of a portfolio over an investor's lifetime and thus seeks to maximize terminal wealth. The origins of the idea can be traced back to the work of Kelly (1956) [2]. According to Estrada (2010) [12], GMM reduces to the following optimization problem

$$\begin{aligned}\mathbf{w}_{gmm} &= \arg \max_{\mathbf{w}} \exp \left[\ln(1 + \mathbf{w}^T \boldsymbol{\mu}) - \frac{\mathbf{w}^T \Sigma \mathbf{w}}{2(1 + \mathbf{w}^T \boldsymbol{\mu})^2} \right] - 1 \\ \text{subject to } & \mathbf{w}^T \mathbf{1} = 1, \mathbf{w}_i \geq 0 \text{ for } i \in \{1, 2, \dots, n\}\end{aligned}$$

where the symbols have their usual meaning.

For the purpose of this study, we employ numerical optimization. We generate 5×10^5 random weight vectors with long-only constraints ($\mathbf{w}^T \mathbf{1} = 1$ and $\mathbf{w}_i \geq 0$ for $i \in \{1, 2, \dots, n\}$), search for the weights vector that

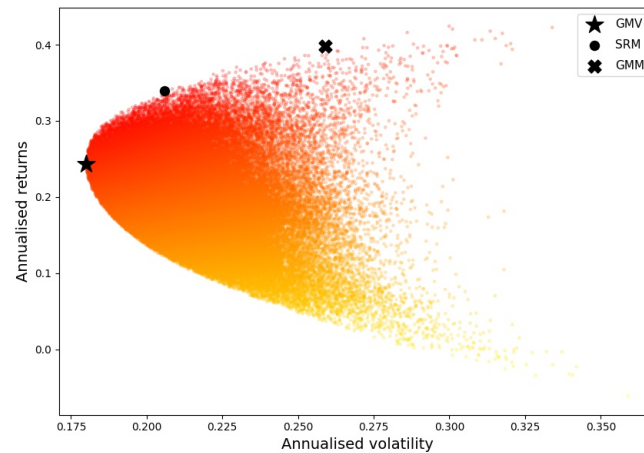


Figure 1: Efficient Frontier showing the Return-Volatility pairs for the three strategies (n=20)

maximize/minimize the required objective functions as above and use them as the final allocation for the particular strategy. This gives us unique weights corresponding to each objective. We plot all the return and volatility pairs on the Efficient Frontier-plot along with the returns and volatility corresponding to three portfolios namely GMV, SRM and GMM, as an illustration in Figure 1.

4 Methodology

Markowitz assumes that μ and Σ are known. However, as pointed out earlier, these parameters need to be estimated from historical data. The estimates thus produced are susceptible to estimation risks. As a result, the allocation is sub-optimal.

Consider the following extreme two-asset example examined by Demiguel et al. (2009) [11] - “Suppose that the true per annum mean and volatility of returns for both assets are the same, 8% and 20%, respectively, and that the correlation is 0.99. In this case, because the two assets are identical, the optimal mean-variance weights for the two assets would be 50%. If, on the other hand, the mean return on the first asset is not known and is estimated to be 9% instead of 8%, then the mean-variance model would recommend a weight of 635% in the first asset and 535% in the second. That is, the optimization tries to exploit even the smallest difference in the two assets by taking extreme long and short positions without taking into account that these differences in returns may be the result of estimation error.”

Our strategy is focused on reducing this estimation error to generate stable weights that give better out-of-sample performance compared to a naive equally-weighted (EW) approach. It is inspired by the machine learning technique of ensemble models. This derives from the knowledge that averaging a set of observations reduces variance. Note that given a set of n independent observations $Z_1, \dots, Z_n$, each with variance σ^2 , the variance of the mean $\bar{Z}$ of the observations is given by σ^2/n .

In a similar vein, we calculate weights for GMV, SRM, GMM and EW portfolios and average them into the ensemble portfolio, hereafter referred to as MV-MIX portfolio.

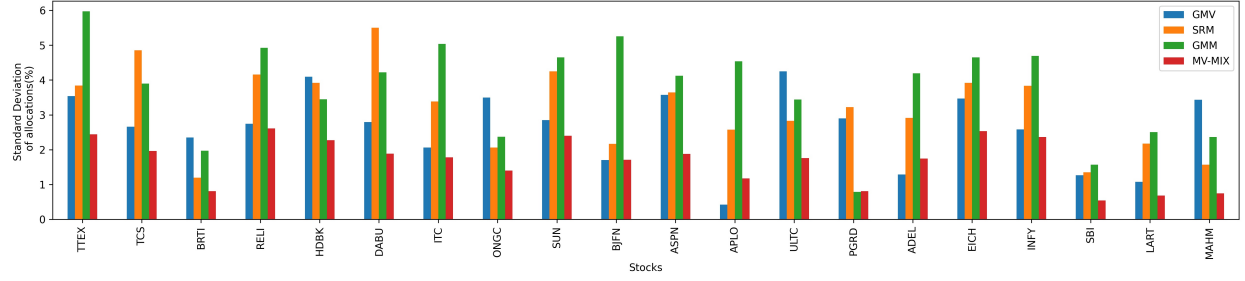


Figure 2: A visual comparison of percentage standard deviation of weights across the 20 assets across all testing sets

Using data on 20 Indian securities listed on the Bombay Stock Exchange (BSE) spanning across 12 years we generate ensemble portfolios based on rolling 2-year look-back periods² and compare the variance in weights against other three strategies in Figure 2. Note the variance of EW portfolio is by construction zero and not shown here. The MV-MIX portfolio weights exhibit least variance for all assets but one. This validates that the ensemble approach leads to a lower variance and hence protects the investor from taking extreme positions as a result of estimation error in $\hat{\mu}$ and $\hat{\Sigma}$.

To test the out-of-sample performance of our strategy, we conduct an empirical analysis using a data set of securities listed on the Bombay Stock Exchange (BSE). We extract the daily price data from January 2008 to December 2019 from Investing.com using an open source python package - *investpy* [19]. Descriptive statistics for the same are provided in Table 1.

Table 1: Descriptive Statistics with first four moments for daily returns

Tickers	Mean	Volatility	Skewness	Excess Kurtosis
TTEX	0.271	0.426	0.942	4.703
TCS	0.258	0.307	0.316	2.435
BRTI	0.066	0.352	0.273	-0.447
RELI	0.123	0.323	-0.044	2.695
HDBK	0.232	0.277	0.580	6.298
DABU	0.238	0.276	0.498	6.939
ITC	0.146	0.267	-0.008	0.703
ONGC	0.012	0.331	0.339	5.661
SUN	0.174	0.315	-0.214	0.932
BJFN	0.607	0.418	1.003	7.058
ASPN	0.311	0.268	0.119	0.607
APLO	0.192	0.429	0.151	3.371
ULTC	0.182	0.309	0.337	0.429
PGRD	0.065	0.284	-0.119	8.059
ADEL	0.247	0.505	0.371	3.776
EICH	0.513	0.382	0.982	6.056
INFY	0.160	0.298	-0.492	9.924
SBI	0.109	0.381	1.610	17.164
LART	0.094	0.351	1.659	22.409
MAHM	0.150	0.354	0.764	8.140

* Mean and volatility have been annualized.

The testing procedure is outlined as below:

1. The price data is converted into daily returns.

²This means that we use two years of data to derive our sample estimates.

2. Training and test sets³ are created in a rolling manner. For example- first we create a training set with daily returns for the years 2008 and 2009; then, the corresponding test set is created with 2010 data. Next, we create training data by shifting a year ahead i.e. using 2009 and 2010 data and testing with 2011 data and so on.
3. Estimate the expected daily returns and the shrunk covariance matrix for each training set.
4. Using these estimates, we calculate portfolio weights for GMV, SRM, GMM and EW portfolios.
5. We ensemble these strategies by calculating the average weight vector.
6. Using the corresponding test data, we calculate the returns on the MV-MIX portfolio.

Following the above procedure, we break the 12-year period into 10 training-testing sets. For the i^{th} training set, $i \in \{1, 2, \dots, 10\}$, we calculate the parameters, $\hat{\mu}_i$ and $\hat{\Sigma}_i$. Using these estimates, we calculate weights w_s^i where $s \in \{GMV, SRM, GMM\}$. We add these weights with the EW weight vector (equal weights) and generate the MV-MIX portfolio. The whole process has been represented pictorially in Figure 7.

Table 2: An out-of-sample comparison of MV-MIX and EW portfolios (n=20)

i^{th} Test set	MV-MIX			EW		
	Sharpe Ratio	Mean Returns	Volatility	Sharpe Ratio	Mean Returns	Volatility
1	0.162	0.431	0.128	0.137	0.397	0.141
2	-0.040	-0.081	0.175	-0.058	-0.130	0.182
3	0.178	0.452	0.122	0.146	0.416	0.138
4	0.075	0.251	0.165	0.067	0.231	0.169
5	0.222	0.609	0.127	0.201	0.556	0.130
6	0.047	0.169	0.173	0.035	0.133	0.176
7	0.021	0.082	0.157	0.020	0.082	0.161
8	0.190	0.436	0.111	0.181	0.397	0.107
9	0.031	0.107	0.152	0.025	0.090	0.148
10	0.041	0.136	0.151	0.026	0.095	0.154

Notes: Mean and volatility have been annualized.

Sharpe Ratio is defined as the expected excess returns over the risk free asset per unit of volatility; $SR = \frac{E[r_i - r_f]}{\sigma}$

In Table 2, we provide a comparison of the ensemble portfolio i.e. MV-MIX and the naive EW portfolio. For the purposes of comparison, we define the *effectiveness index* (EI) as the number of test sets where the MV-MIX portfolio has outperformed the EW portfolio over the total number of test sets for the selected metric. For $n = 20$, the effectiveness index with respect to Sharpe ratio is 1⁴. We can clearly see that the MV-MIX portfolio has outperformed the naive EW portfolio. We check the robustness of these results in the subsection 4.1.

³The training set here refers to the look-back period used to estimate sample mean vector and sample covariance matrix which are used for optimizing the selected objective function to get the weights and the testing period refers to the holding period for which the performance of the optimized weights is checked.

⁴As for all the test sets MV-MIX displays higher Sharpe ratios compared to EW portfolios

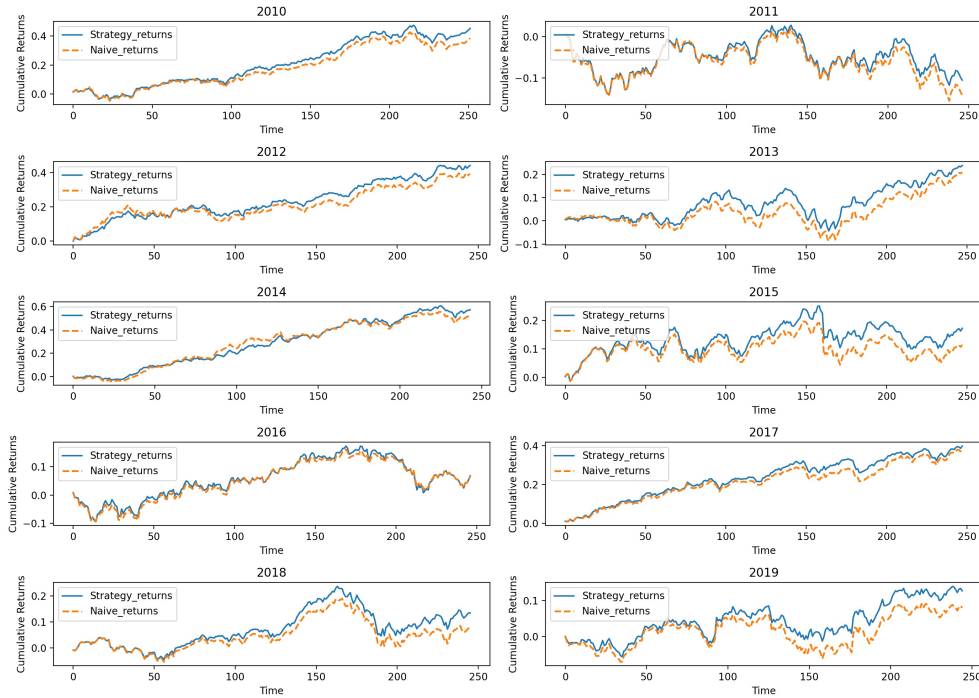


Figure 3: A comparison of out-of-sample Cumulative returns of MV-MIX and EW portfolio across the 10 test periods

Figure 3 depicts a comparison of the MV-MIX and EW portfolio cumulative returns. We can see that across the test periods, the MV-MIX terminal wealth is greater than or equal to that of the EW portfolio. Figure 4 depicts the performance of the two strategies over the entire 10-year testing period with yearly re-balancing. Along with a large out-performance in terms of terminal wealth, the MV-MIX portfolio also exhibits a slightly higher Sharpe ratio of 1.29 compared to 1.26 for EW. Further, Figure 5 shows the comparison between drawdowns for the two strategies. While terminal wealth and Sharpe ratio are metrics focused on returns/risk-adjusted returns, drawdown is a measure of downside volatility. MV-MIX portfolio generally exhibits smaller drawdowns in comparison to EW portfolio.



Figure 4: Cumulative returns comparison for the entire dataset with yearly rebalancing

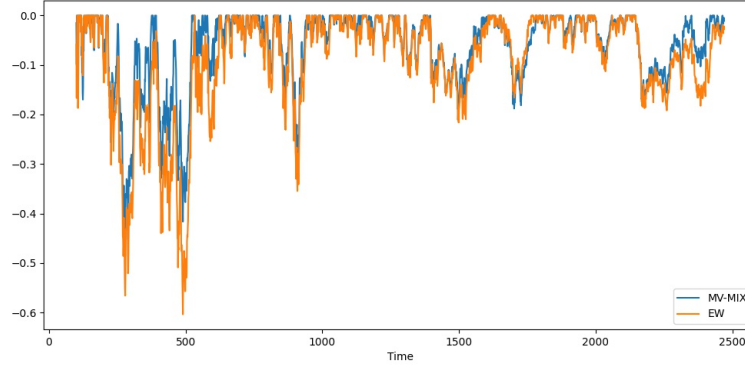


Figure 5: Drawdown comparison of the two strategies across the entire period

4.1 Robustness checks

To validate the results observed for the above case, we run similar experiments under varying conditions. We check the sensitivity of the effectiveness index under three scenarios by varying:

1. Number of assets (n)
2. Covariance matrix estimators
3. Size of the training set or the look-back period

We provide the results for the same in Table 4 through 11 in the Appendix. The results from these tables have been summarized in Table 3 using the Effectiveness Index on the basis of Sharpe Ratio.

Table 3: Robustness checks using Effectiveness Index (EI)

	Number of assets (n)				Covariance estimator			Size: training set (years)		
	20	15	10	5	Bayes Wishart	Ledoit-Wolf	Sample covar	5	3	1
EI (Sharpe Ratio)	1.0	1.0	1.0	0.6	0.8	1.0	0.9	0.85	0.88	0.77

As a first test of robustness, we vary the number of assets in the portfolio to test if the MV-MIX portfolio's superiority over EW is robust. We select random subset of assets from the original set of 20. As we can see, for $n \in \{5, 10, 15, 20\}$, EI is greater than 0.5. As the number of assets for the case of $n = 5$ is low, there is lesser chance of distributing the weights across assets and as a result the MV-MIX portfolio is comparatively less effective compared to EW portfolio.

Secondly, we test whether a particular estimator of covariance is better suited to this approach. We compare the performance of Bayesian shrinkage under Wishart prior, the linear shrinkage approach under constant variance constraints (Ledoit-Wolf 2004) and the sample covariance matrix. We find invariance in performance across these estimators. However, the Ledoit-Wolf estimator is better compared to the rest of the approaches.

Lastly, we look at whether larger or smaller look-back periods have any effect on the strategy. Instead of using a two-year look-back period as in Section 5, we test for 5 years, 3 years and 1 year training periods and test on 1-year ahead out-of-sample returns. Again, we find the MV-MIX portfolio to be robust to the size of the training period⁵.

⁵Note that changing the size of the training set changes the number of training-testing pairs available. For a size of 5 yrs, we have 7 rolling train-test pairs and similarly for 3 yrs and 1 yrs we have 9 and 11 pairs respectively.

5 Conclusion

Individual and institutional investors alike are faced with the problem of efficient portfolio allocation. The mean-variance approach of Markowitz provides a sound and powerful solution to this problem. However, in practice it results in sub-optimal allocations. The primary culprits discussed in the literature are poor estimates of expected returns and the covariance matrix. Demiguel et al. (2009) find that due to estimation errors, Mean-Variance Optimal portfolios find it hard to beat the naive equally-weighted portfolios.

Literature suggests that much emphasis has been on reducing the estimation error at source, by introducing approaches aimed at recovering better estimates of input parameters. However, we believe that even after using improved estimates, we run the risk of overfitting on the historical data by employing only one strategy. Therefore, we introduce an ensemble-inspired approach where we combine three optimal strategies with the equally-weighted strategy. This ensemble strategy reduces the risk of overfitting by pulling extreme positions to an average value. Additionally, it reduces the estimation risk by bounding the variance in the allocations. The approach is tested using an empirical study and the results are presented. Robustness checks are conducted to validate the performance persistence of the strategy.

Our approach opens up a few interesting research pathways in portfolio optimization. Future research may include treating the weight allocation across strategies as an optimization problem in itself. The approach can be combined with the Black-Litterman and similar approaches that put structure on the expected returns as well and check whether it further improves the strategy. Finally, the idea merits a comprehensive empirical study involving data from several market universes for further validation. Performance can also be tested against more advanced asset allocation strategies for example the Hierarchical Risk Parity (HRP) approach.

A Appendix

A.1 Tables and Figures

Here we provide the supporting tables and figures as discussed in the main text. Mean returns and volatility has been annualized for all the tables.

Table 4: An out-of-sample comparison of MV-MIX and EW portfolios (n=15)

i^{th} Test set	MV-MIX			EW		
	Sharpe Ratio	Mean Returns	Volatility	Sharpe Ratio	Mean Returns	Volatility
1	0.187	0.544	0.137	0.152	0.451	0.143
2	-0.034	-0.064	0.175	-0.057	-0.131	0.186
3	0.208	0.562	0.127	0.147	0.445	0.146
4	0.091	0.305	0.165	0.072	0.243	0.167
5	0.250	0.691	0.125	0.219	0.615	0.130
6	0.026	0.104	0.172	0.015	0.071	0.172
7	0.040	0.148	0.173	0.028	0.107	0.166
8	0.171	0.362	0.104	0.168	0.386	0.112
9	0.058	0.192	0.160	0.049	0.161	0.155
10	0.055	0.173	0.152	0.024	0.090	0.155

Table 5: An out-of-sample comparison of MV-MIX and EW portfolios (n=10)

i^{th} Test set	MV-MIX			EW		
	Sharpe Ratio	Mean Returns	Volatility	Sharpe Ratio	Mean Returns	Volatility
1	0.130	0.419	0.156	0.123	0.433	0.170
2	-0.032	-0.060	0.178	-0.060	-0.149	0.198
3	0.195	0.652	0.154	0.149	0.503	0.161
4	0.086	0.311	0.178	0.078	0.282	0.177
5	0.234	0.939	0.171	0.212	0.732	0.156
6	0.063	0.253	0.198	0.056	0.231	0.203
7	0.023	0.106	0.198	0.022	0.099	0.189
8	0.184	0.538	0.138	0.168	0.464	0.132
9	0.022	0.095	0.182	0.006	0.045	0.171
10	0.053	0.195	0.178	0.026	0.108	0.179

Table 6: An out-of-sample comparison of MV-MIX and EW portfolios (n=5)

i^{th} Test set	MV-MIX			EW		
	Sharpe Ratio	Mean Returns	Volatility	Sharpe Ratio	Mean Returns	Volatility
1	0.150	0.649	0.199	0.129	0.533	0.196
2	-0.007	0.004	0.199	-0.021	-0.038	0.201
3	0.212	0.719	0.153	0.192	0.613	0.148
4	0.107	0.430	0.195	0.111	0.418	0.183
5	0.221	1.022	0.193	0.200	0.778	0.173
6	0.075	0.382	0.249	0.092	0.468	0.245
7	-0.008	0.000	0.215	-0.020	-0.034	0.196
8	0.105	0.349	0.164	0.093	0.286	0.151
9	-0.044	-0.105	0.196	-0.030	-0.058	0.182
10	-0.056	-0.153	0.218	-0.031	-0.076	0.213

Table 7: An out-of-sample comparison of MV-MIX and EW portfolios: Ledoit-Wolf (2004) estimator (n=20)

i^{th} Test set	MV-MIX			EW		
	Sharpe Ratio	Mean Returns	Volatility	Sharpe Ratio	Mean Returns	Volatility
1	0.166	0.446	0.130	0.137	0.397	0.141
2	-0.041	-0.081	0.172	-0.058	-0.130	0.182
3	0.191	0.479	0.120	0.146	0.416	0.138
4	0.072	0.240	0.165	0.067	0.231	0.169
5	0.216	0.573	0.124	0.201	0.556	0.130
6	0.051	0.192	0.185	0.035	0.133	0.176
7	0.023	0.090	0.161	0.020	0.082	0.161
8	0.182	0.386	0.104	0.181	0.397	0.107
9	0.023	0.086	0.150	0.025	0.090	0.148
10	0.037	0.123	0.152	0.026	0.095	0.154

Table 8: An out-of-sample comparison of MV-MIX and EW portfolios: Sample Covariance estimator (n=20)

i^{th} Test set	MV-MIX			EW		
	Sharpe Ratio	Mean Returns	Volatility	Sharpe Ratio	Mean Returns	Volatility
1	0.155	0.446	0.139	0.137	0.397	0.141
2	-0.046	-0.097	0.177	-0.058	-0.130	0.182
3	0.182	0.465	0.123	0.146	0.416	0.138
4	0.070	0.240	0.168	0.067	0.231	0.169
5	0.235	0.654	0.128	0.201	0.556	0.130
6	0.043	0.163	0.181	0.035	0.133	0.176
7	0.007	0.045	0.158	0.020	0.082	0.161
8	0.206	0.438	0.102	0.181	0.397	0.107
9	0.042	0.137	0.150	0.025	0.090	0.148
10	0.047	0.151	0.151	0.026	0.095	0.154

Table 9: An out-of-sample comparison of MV-MIX and EW portfolios (n=20, look-back period = 5 years)

i^{th} Test set	MV-MIX			EW		
	Sharpe Ratio	Mean Returns	Volatility	Sharpe Ratio	Mean Returns	Volatility
1	0.077	0.261	0.166	0.067	0.231	0.169
2	0.234	0.635	0.125	0.201	0.556	0.130
3	0.034	0.124	0.164	0.035	0.133	0.176
4	0.020	0.082	0.160	0.020	0.082	0.161
5	0.188	0.399	0.103	0.181	0.397	0.107
6	0.030	0.101	0.145	0.025	0.090	0.148
7	0.039	0.129	0.150	0.026	0.095	0.154

Table 10: An out-of-sample comparison of MV-MIX and EW portfolios (n=20, look-back period = 3 years)

i^{th} Test set	MV-MIX			EW		
	Sharpe Ratio	Mean Returns	Volatility	Sharpe Ratio	Mean Returns	Volatility
1	-0.036	-0.067	0.170	-0.058	-0.130	0.182
2	0.182	0.470	0.124	0.146	0.416	0.138
3	0.075	0.247	0.161	0.067	0.231	0.169
4	0.236	0.629	0.123	0.201	0.556	0.130
5	0.047	0.175	0.178	0.035	0.133	0.176
6	0.016	0.070	0.158	0.020	0.082	0.161
7	0.193	0.417	0.105	0.181	0.397	0.107
8	0.039	0.125	0.145	0.025	0.090	0.148
9	0.041	0.134	0.150	0.026	0.095	0.154

Table 11: An out-of-sample comparison of MV-MIX and EW portfolios (n=20, look-back period = 1 years)

i^{th} Test set	MV-MIX			EW		
	Sharpe Ratio	Mean Returns	Volatility	Sharpe Ratio	Mean Returns	Volatility
1	0.202	1.166	0.232	0.202	1.453	0.272
2	0.160	0.471	0.142	0.137	0.397	0.141
3	-0.045	-0.087	0.168	-0.058	-0.130	0.182
4	0.163	0.415	0.124	0.146	0.416	0.138
5	0.081	0.266	0.163	0.067	0.231	0.169
6	0.202	0.517	0.121	0.201	0.556	0.130
7	0.033	0.122	0.168	0.035	0.133	0.176
8	0.017	0.074	0.161	0.020	0.082	0.161
9	0.188	0.379	0.099	0.181	0.397	0.107
10	0.031	0.105	0.148	0.025	0.090	0.148
11	0.039	0.120	0.139	0.026	0.095	0.154

A.2 Drawdown

Figure 6 shows a visual comparison of drawdown between two strategies across the test periods.

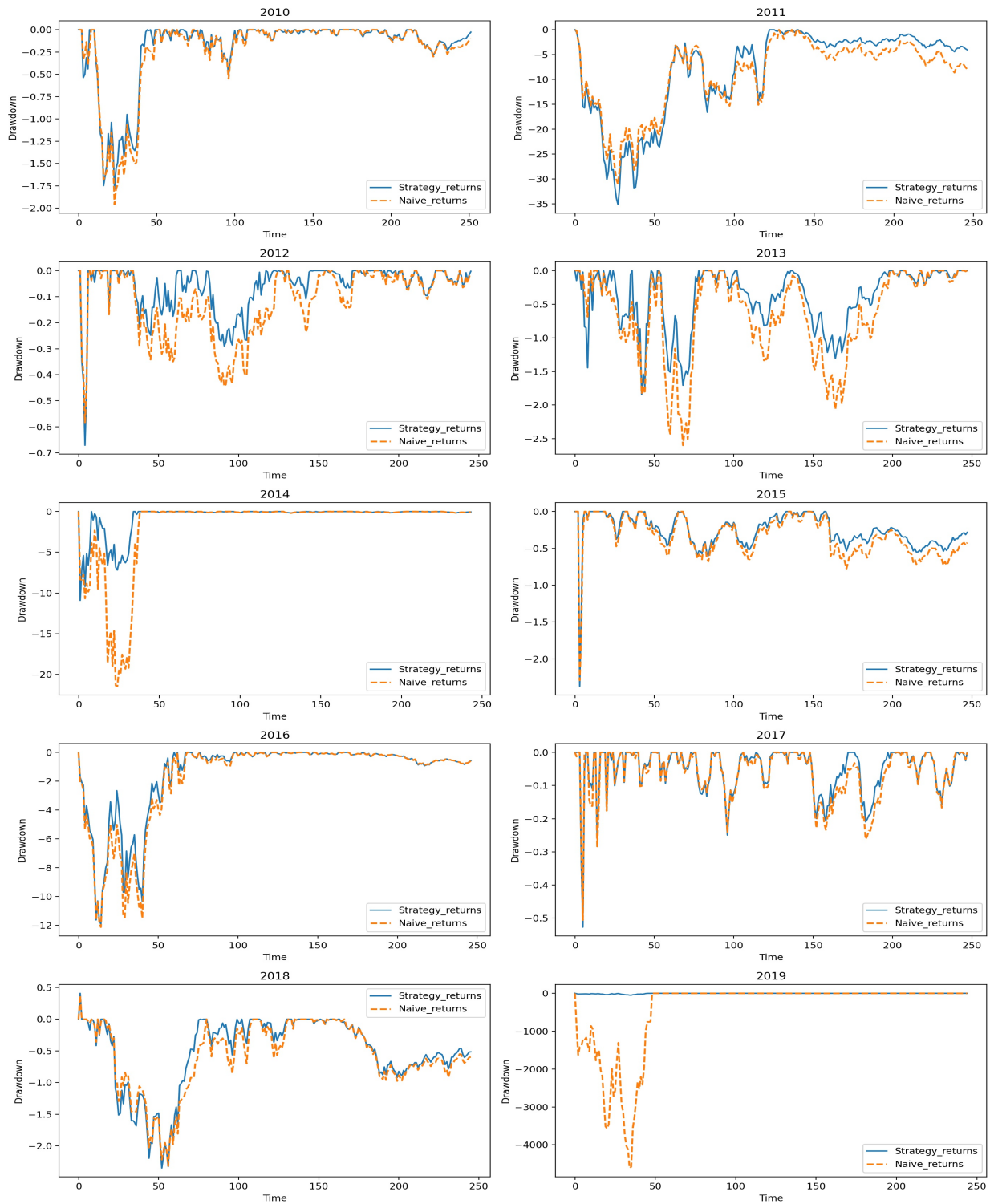


Figure 6: Drawdown comparison across the testing period

A.3 Ensemble approach

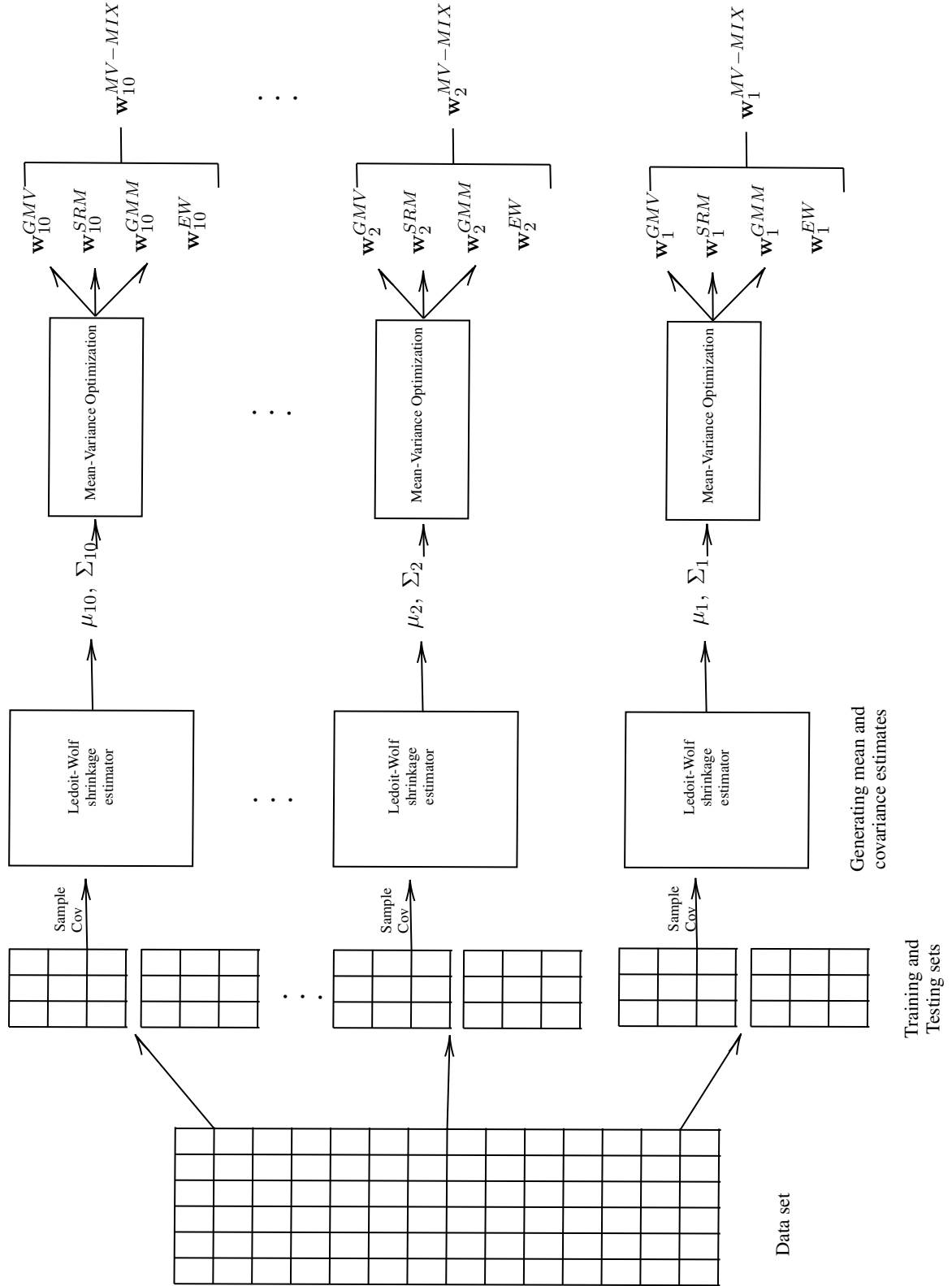


Figure 7: The ensemble approach

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